

Exemplary tasks related to Data Converters - Summer Term 2026

1. A/D Conversion

- 1.1. Sketch the signal processing chain of an A/D conversion.
- 1.2. Briefly explain the functionality of each block.

2. D/A Conversion

- 2.1. Sketch the signal processing chain of a D/A conversion.
- 2.2. Briefly explain the functionality of each block.

3. Aliasing

A sine-wave signal with an amplitude of 0.2 mV and a frequency of 15 kHz is sampled with a sampling frequency of 100 kHz. Interferers are located at 130 kHz (amplitude 0.5 V), 275 kHz (Amplitude 0.3 V), and 315 kHz (amplitude 0.4 mV).

- 3.1. Sketch the single-sided FFT (amplitude vs. frequency) before the sampling is performed.
- 3.2. Sketch the single-sided FFT (amplitude vs. frequency) after the sampling is performed.

4. Quantization

- 4.1. Draw the linear model of a quantizer with resolution B and full-scale FS .
- 4.2. How large is the quantization error?
- 4.3. Briefly explain the quantizer characteristics unipolar, bipolar, mid-rise and midtread.
- 4.4. How large is the power of the quantization error?
- 4.5. Sketch and label the power spectral density plot (PSD) of the quantization error.
- 4.6. Derive the signal-to-quantization-noise ratio (SQNR) of the quantizer.
- 4.7. By how many dB increases the SQNR if the resolution increases by 21 dB?

5. Offset & gain Error

- 5.1. Briefly explain offset and gain error of an ADC and sketch the corresponding transfer characteristics in comparison with the ideal transfer characteristic.
- 5.2. Briefly explain offset and gain error of a DAC and sketch the corresponding transfer characteristics in comparison with the ideal transfer characteristic.

6. DNL & INL

- 6.1. Briefly explain DNL and INL of an ADC.
- 6.2. How large is the DNL of the first and last code of any ADC?
- 6.3. How large is the INL of the first code of any ADC?
- 6.4. How large is the INL of the second and last code of an ADC?
- 6.5. Briefly explain DNL and INL of a DAC.
- 6.6. Sketch and explain why the monotonicity of a DAC can be guaranteed if $DNL < 1 \text{ LSB}$.

7. Spectral Metrics

- 7.1. Briefly explain the characteristics SQNR, SNR, SNDR, SFDR, and THD of an ADC.

8. DAC

- 8.1. All DACs can be categorized into three general types. State the three types.
 - Derive and compare the number of unit elements which are required for the implementation of a 6-bit DAC of each type!
 - State advantages / disadvantages of each type!
- 8.2. Of how many (unit) current sources and switches consists a
 - 4-bit thermometer current steering DAC
 - 4-bit binary weighted current steering DAC
 - 4-bit segmented current steering DAC
- 8.3. Choose your favourite DAC architecture from the selection “resistor string, R-2R ladder, charge scaling”.
 - Draw the schematic for an implementation with a resolution of 2bits.
 - Briefly explain the functionality.
- 8.4. Sketch the block diagram of a cyclic 4-bit DAC!
 - How many clock cycles are needed in order to perform a single conversion?
 - Briefly explain its functionality for a digital input code of 1011.

9. Sampling

- 9.1. Choose your favourite sampler nonideality from the selection “finite acquisition time, kT/C noise, aperture uncertainty, signal dependent sampling instant,

feedthrough & drop mode, track mode nonlinearity, and charge injection” and briefly explain the nonideality.

- 9.2. Sketch and explain “bootstrapping”.
- 9.3. Sketch and explain “charge cancellation”.
- 9.4. Sketch and explain “bottom plate sampling”.

10. Comparators

- 10.1. Which circuit blocks are typically used for the implementation of a comparator?
- 10.2. Sketch “output series cancellation” and derive the transfer function.
- 10.3. Sketch “input series cancellation” and derive the transfer function.

11. ADCs

- 11.1. Sketch and explain the functionality of a single-slope ADC.
- 11.2. Sketch and explain the functionality of a dual-slope ADC.

12. Delta-Sigma Modulator

- 12.1. Sketch the block diagram of a first-order discrete-time Delta-Sigma modulator.
- 12.2. Replace the quantizer by a simple linear model and derive the signal as well as the noise transfer function, i.e., STF and NTF, respectively.
- 12.3. Name and explain the two principles applied to a Delta-Sigma modulator in order to implement a high-resolution A/D converter although only a low-resolution A/D converter (quantizer) is used in a Delta-Sigma A/D converter